HyperLocalize

Context-Aware Identity-Preserving Video Localization

[STUDENT NAME] · [REGISTER NUMBER]
Guide: [GUIDE NAME], [DESIGNATION]
[DEPARTMENT OF CSE] · [COLLEGE NAME] · [MONTH YEAR]

Automatic dubbing treats AI output as final



Trust

A dropped "not" or a changed number (2 sachets → 5) can reverse a health message, and nobody is asked to check.



Identity

Voice cloning can copy a real speaker. Consent is at best a one-time tick box, and revoking it rarely stops work in progress.



Accountability

The output file carries no record of who reviewed it, which models made it, or which consent applied.

Problem: build a localization workflow where AI output is only a draft, critical content is checked, a human approves every scene, identity is used only under valid consent, and every action is auditable.

A controlled workflow, not an autonomous pipeline

HyperLocalize is

- A general, context-aware video localization workflow
- An information-access layer for important video information (health, food, water, energy)
- Human-reviewed: AI output is always a draft
- Consent-governed for any use of a speaker's identity
- Traceable: every governance action is audited

HyperLocalize is not

- A health, ORS or Tamil-only app (that is the demo scenario)
- A voice-cloning product
- A competitor on raw dubbing quality
- A fully autonomous pipeline
- A solution to physical problems in any domain
- Described as "secure", "compliant" or "production-ready"

Objectives

Context-aware drafts

Localize each line with its neighbours, speaker, audience, glossary and timing budget

Critical-content checks

Numbers, units, glossary, structure and overrun checks block approval

Scene-level regeneration

Redo one problematic scene without touching the rest

Enforced consent

Backend gate at every stage; revocation stops running jobs

Approval and export gates

Frozen snapshot, separation of duties, MOCK never exported

Auditable evidence

Hash-chained audit log; chain head inside every export

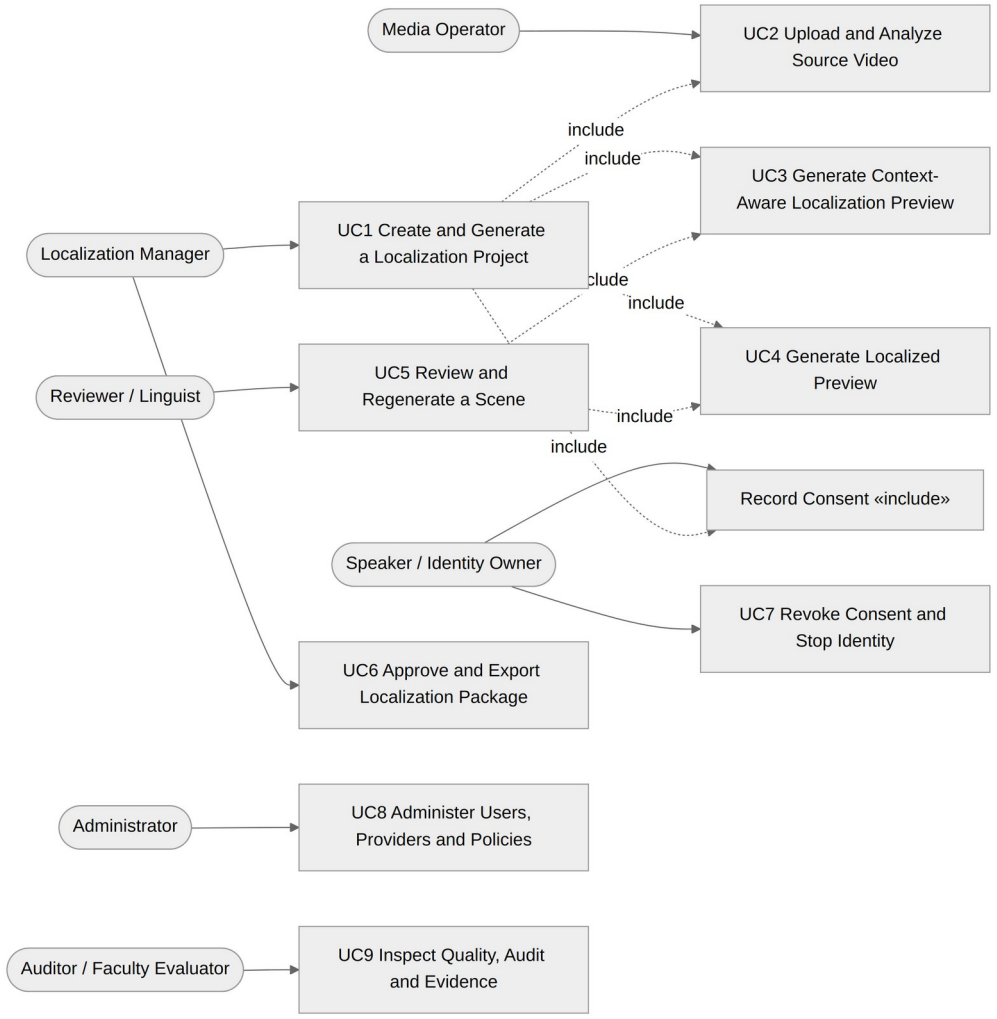
The workflow: AI drafts, humans decide



Human governance steps: only a person can accept, approve or export

Asynchronous jobs: never block a web request

Six actors, nine use cases



Localization Manager
Creates projects, approves, exports

Reviewer / Linguist
Edits, overrides with reasons, regenerates, accepts scenes

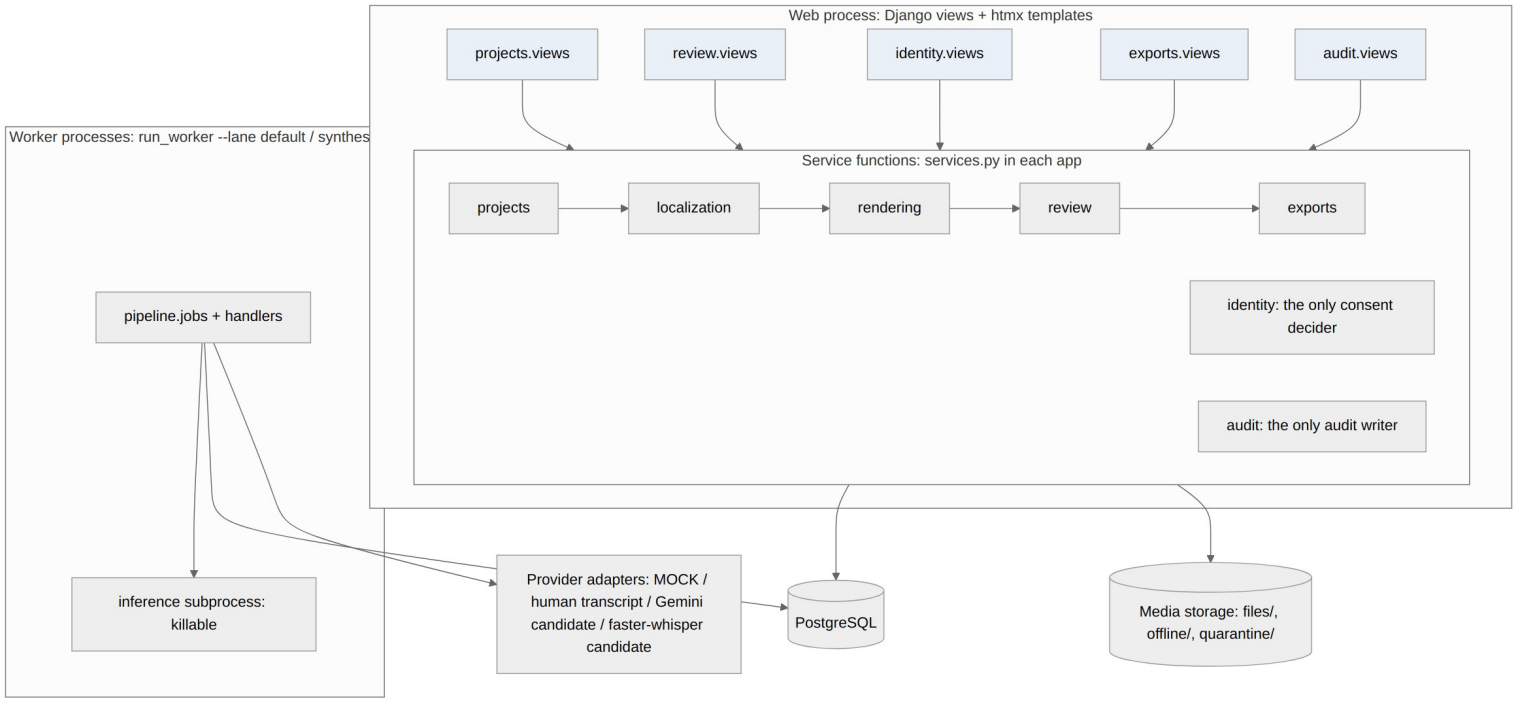
Media Operator
Uploads video, links speakers, watches jobs

Speaker / Identity Owner
Grants, narrows, revokes or withdraws consent

Administrator
Users, provider chains, policies, quota limits (cannot grant consent)

Auditor / Faculty Evaluator
Inspects audit trail, verifies the hash chain

Architecture: a modular monolith



Web never blocks

Long work becomes jobs in a hand-written PostgreSQL queue (SKIP LOCKED)

Killable inference

Models run in a subprocess the worker can kill on revocation

9 Django apps

Apps call each other only through services.py; no signals

Why not microservices? One developer, a laptop demo and a ₹0 budget; database locks give consent the guarantees it needs (ADR 0001).

Technology stack (₹0 budget)

Django 6.1.1

Backend, no REST framework

PostgreSQL 17

Data, job queue, audit triggers

Templates + htmx

Server-rendered pages per role

ffmpeg / ffprobe

Validation, shot cuts, mixing

faster-whisper small

English ASR on CPU (candidate)

Gemini free tier

Context-aware translation (candidate)

MMS-TTS Tamil

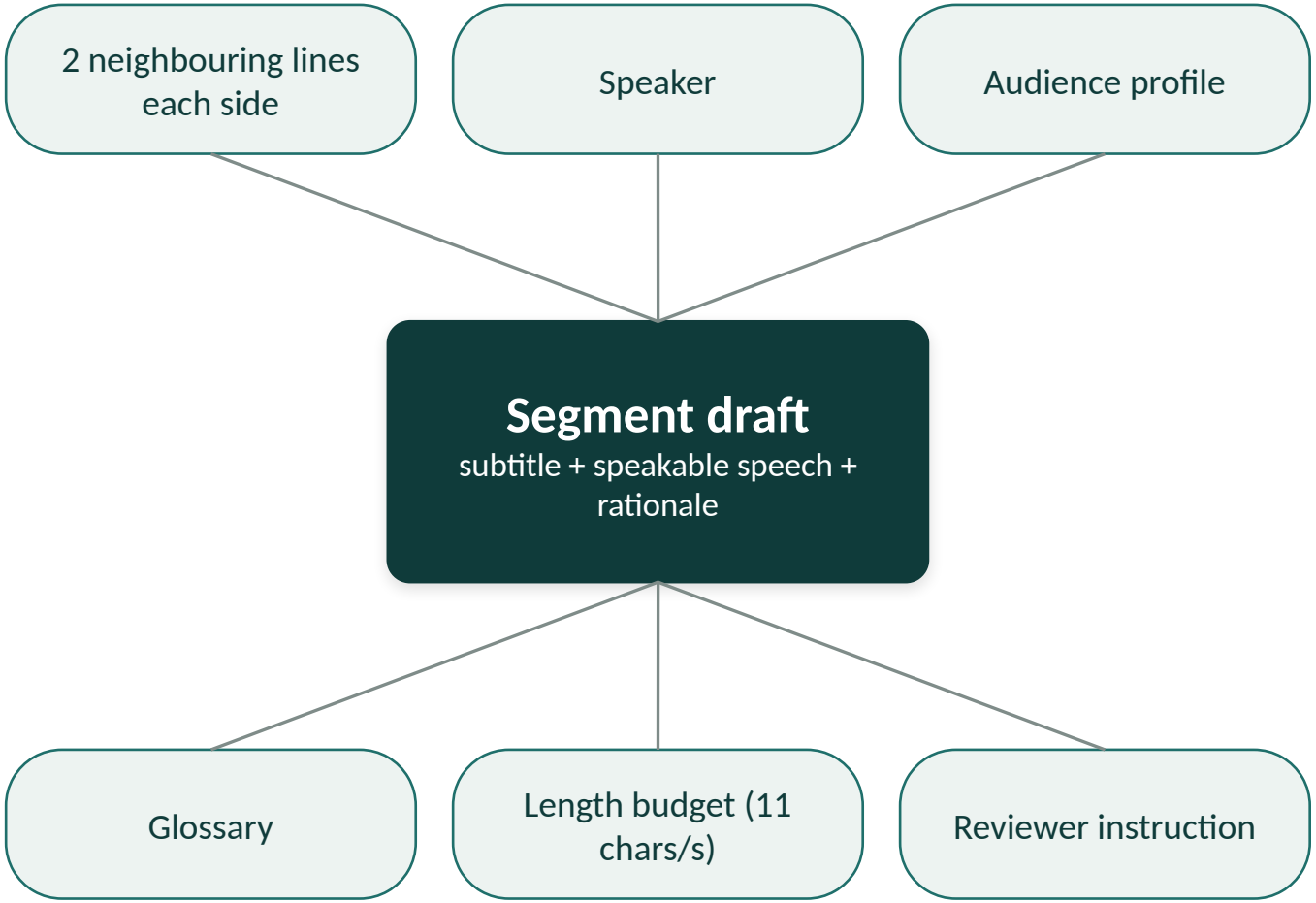
Stock voice on CPU (non-commercial licence)

Docker Compose

Files written; not yet run

Marigold = AI candidates behind provider interfaces; final choices wait for the measured evaluation (spike).

Context-aware localization



Quota-safe by design

- Free tier: about 20 requests / day / model (measured)
- Whole clip in 1 request, meaning check in 1 more (was ~16 for clip 01)
- Provider fails → scenes stay PENDING, "Translation pending" + Retry
- Never silently fills in MOCK text
- Every draft records its context, adapter, model and prompt version

Visual context is not used yet. Whether context improves quality is NOT MEASURED.

Critical-content checks block approval

Check	What it catches	Effect
Numbers	Changed or missing numbers; Tamil digits and number words 0–100	FAIL / UNVERIFIABLE
Units	Missing or changed units; declared conversions	FAIL / UNVERIFIABLE
Glossary	Key terms not using an accepted target form (prefix match for inflections)	No forms → UNVERIFIABLE, never PASS
Structure	Empty, untranslated, mostly Latin script, over reading budget	FAIL
Speech overrun	Line still too long at the 1.2× speed-up cap	FAIL: "shorten the speech"
AI meaning flag	Possible meaning reversal, e.g. a dropped "not"	Needs acknowledgement



A FAIL or UNVERIFIABLE result blocks approval until the text is fixed or a reviewer overrides it with a written, audited reason. Known gap: numbers are compared as counts, not positions.

Consent is enforced in the backend

Identity use is allowed only if

- the owner has not withdrawn
- project consent is ACTIVE, unexpired, covers the use and language
- it sits within an ACTIVE standing consent for this purpose
- the job is not flagged for cancellation
- re-checked at every stage boundary (checkpoints CP1–CP10), under a row lock on save
- only the owner can grant; an Administrator's grant is refused

On revocation

1. Running job: worker polls every ~1 s and kills the subprocess
2. Scenes → NEEDS_REGENERATION; render jobs cancelled
3. This owner's audio quarantined, then purged
4. Only approvals and exports with this voice invalidated
5. Stock-voice fallback queued; must be reviewed again

< 8 s mid-job stop test bound (actual stop time not recorded)

Three gates before anything leaves the system

Review

- Edits create new HUMAN versions
- Override only with a written reason
- AI flags need acknowledgement
- Accept only with no open checks and a current preview
- Regenerate one scene, with an instruction

Approve

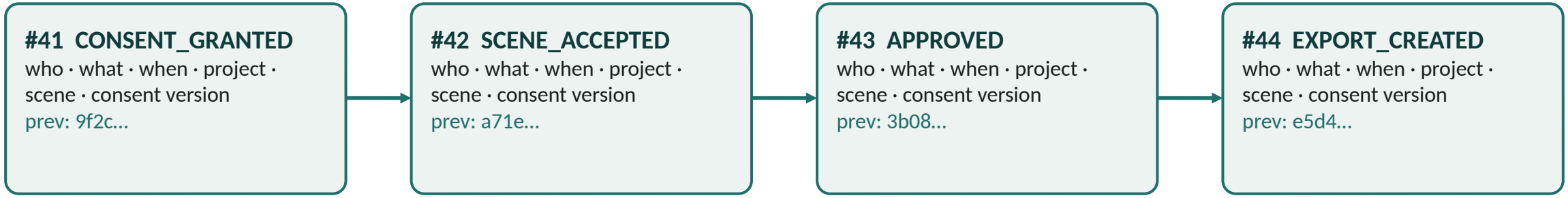
- Every scene accepted
- No blocking checks on latest versions
- Consent CP8 passes
- Separation of duties: approver did not edit
- Freezes an immutable snapshot

Export

- Refused if any artefact is MOCK
- Refused if quarantined or CP9 fails
- MP4 + SRT + VTT (tagged Tamil)
- script.json, quality_report.json
- evidence_manifest.json with chain head

Any edit, regeneration or revocation after approval invalidates it. Exports already downloaded cannot be recalled, and the system says so.

Audit: an append-only hash chain



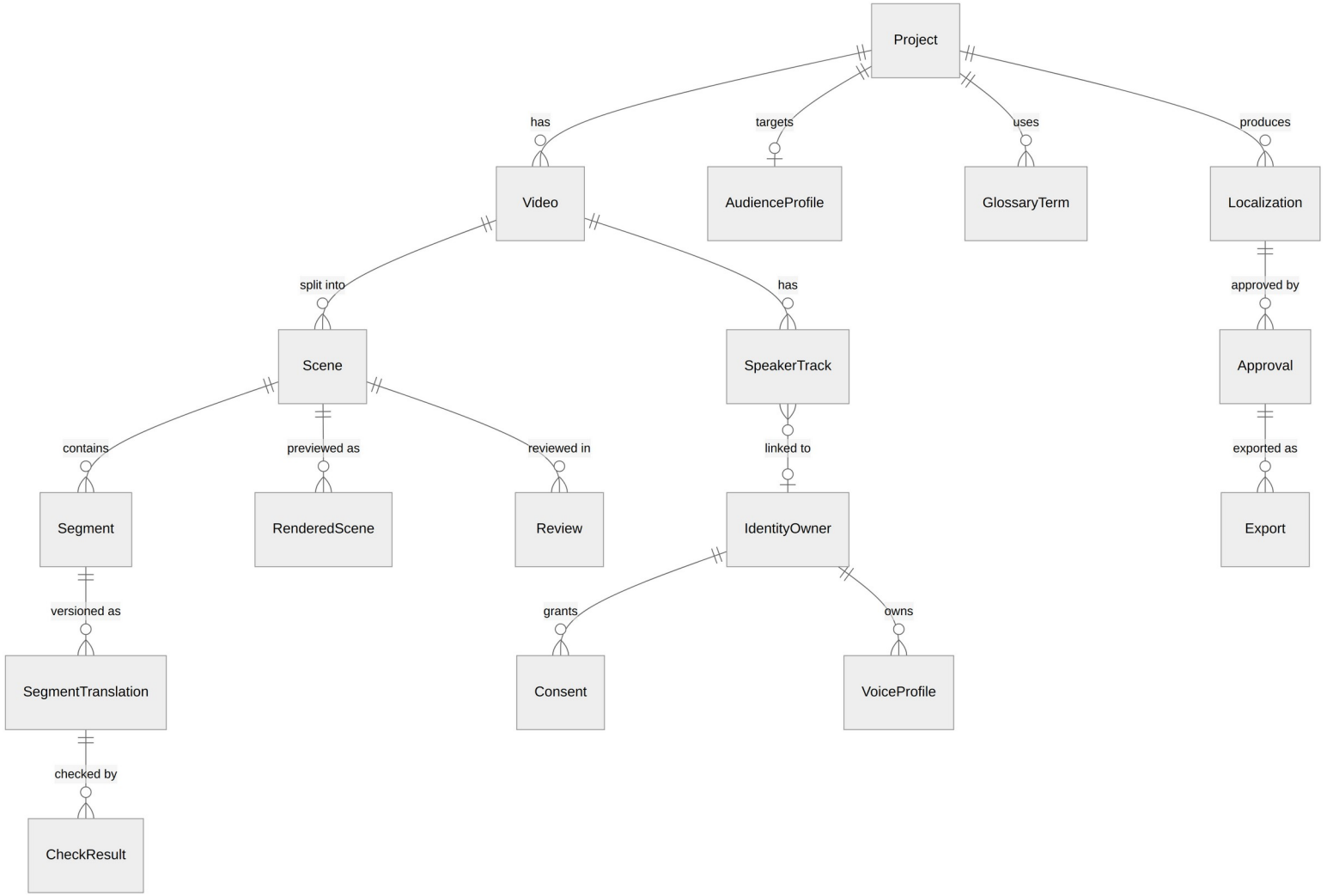
Illustrative records; hash values shortened

```
hash = SHA-256( prev_hash + canonical_json(record) )
```

- One writer (audit.services.record); appends serialised by locking one chain-head row
- PostgreSQL triggers reject UPDATE, DELETE and TRUNCATE
- verify_audit_chain recomputes every hash; each export manifest carries the chain head

Tamper-evident, not tamper-proof

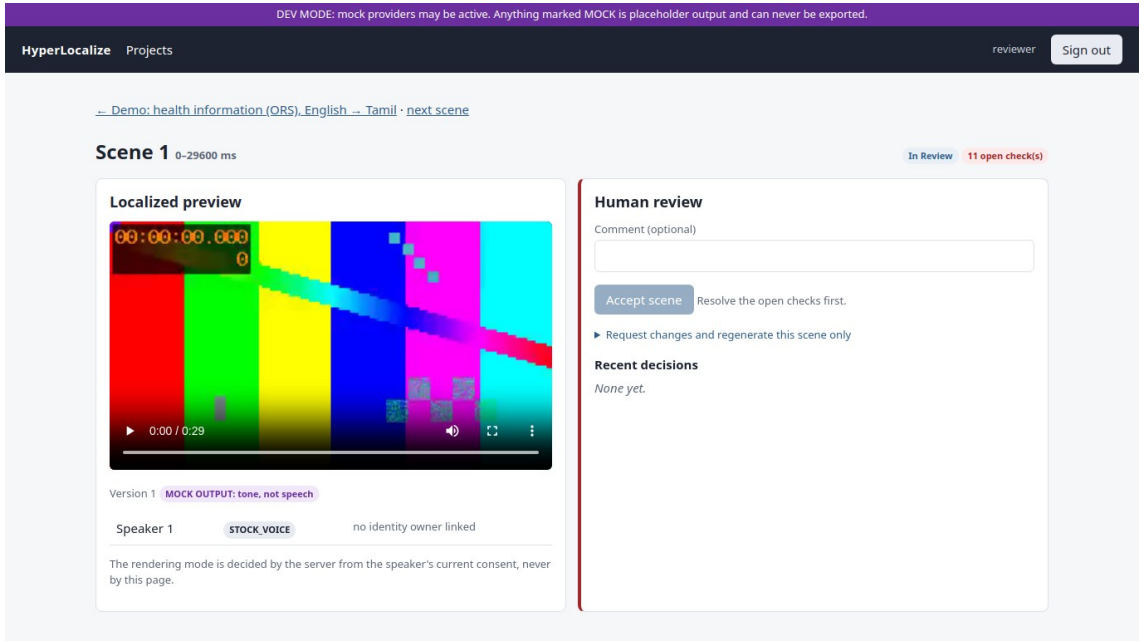
Data model: 28 classes in 9 apps



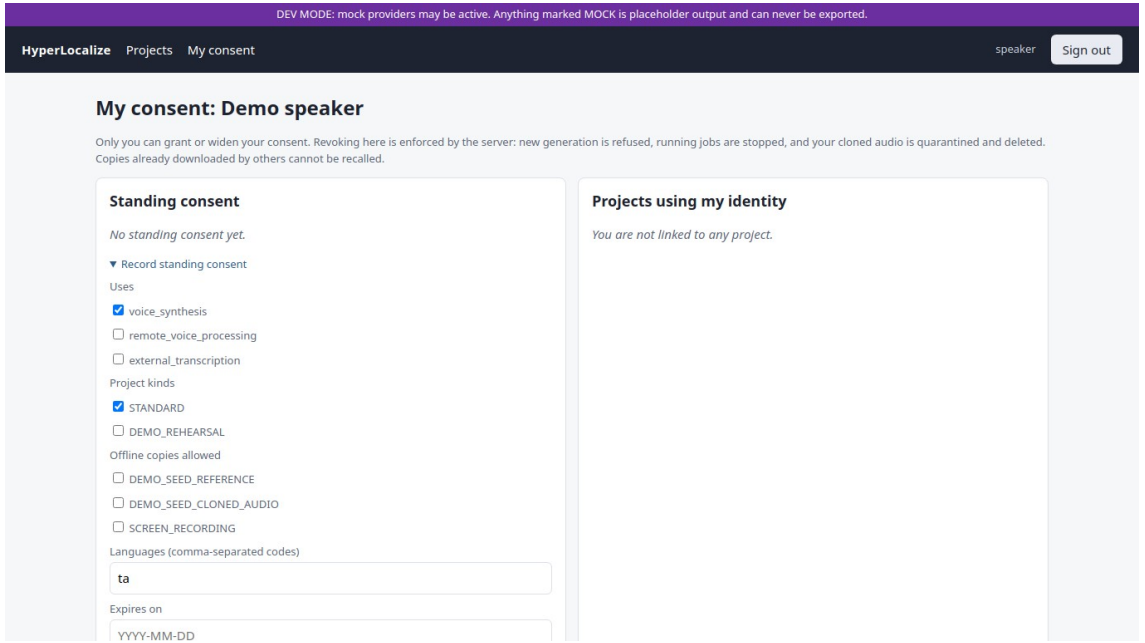
Key invariants

- Translations, consent versions and audit records are append-only
- An Approval stores a snapshot of exact version ids
- Every AI artefact records adapter, model, prompt version and is_mock
- The full class diagram is generated from the Django models

One page per role



Reviewer: preview, per-line checks, override with reason, regenerate this scene only



Speaker / Identity Owner: standing and project consent, reference clip, revoke, withdraw

Screenshots from a seeded test database (2026-10-12). The purple banner and MOCK badges are deliberate: placeholder output is always labelled.

Testing: negative-first, all measured

210

tests passed

2 optional real-model tests skipped · 125 s · 2026-10-22

15/15

end-to-end HTTP steps

live server + worker · 2026-10-06

11

black-box tests passed

equivalence partitions + boundary values

1 + 6

review findings fixed

1 blocker, 6 majors in consent and audit code

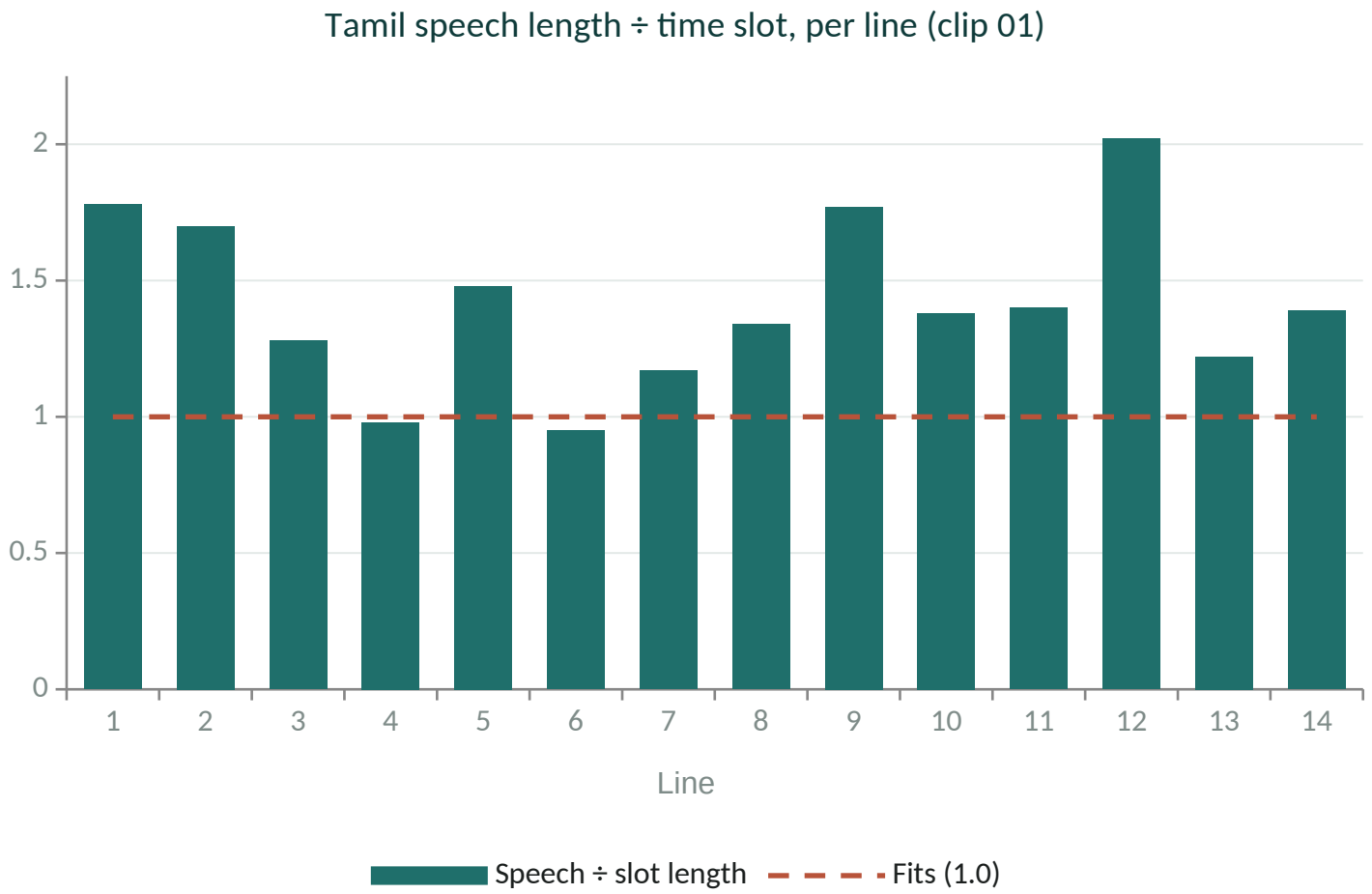
What the tests prove

- Every tested denied consent state is refused; a real subprocess is killed mid-job
- 0 bypasses of the review and approval gates in the tested cases; MOCK never exportable
- Audit edits rejected by the database; tampering detected by the verifier

Not tested yet

- Real models at evaluation scale
- Docker images
- Browser form submission
- Performance and load

First real clip: a trial that found real problems



14 / 14

lines translated by Gemini (after HTTP 429 and 503 fallbacks)

2 / 14

lines fitted their slot: 79.1 s of speech for 58.0 s

0.16

back-transcription CER (intelligibility PROXY, not a listening test)

MEASURED on n = 1 public clip (58 s, CC BY 3.0). Directional, not a quality evaluation.

What the trial changed (ADR 0022)

Problem observed	Change made and tested
Free-tier quota ran out; chain silently fell back to MOCK	Batch whole clip: ~16 → 2 requests; never fall back to MOCK after a real failure; "translation pending" + Retry
Only 2 / 14 lines fitted their slot	Length budget in the prompt (11 chars/s); speed-up cap 1.2×; blocking overrun check
Export tracks tagged "und"	Audio and subtitles tagged with the target language (tam)
Tamil number words unverifiable	Draft parser for 0–100: a wrong number word now FAILS

Re-run (project 5) stopped honestly at "translation pending": daily quota used, 0 translations, 0 MOCK. The timing fix is NOT yet re-measured.

Honest status of the nine use cases

Use case	Status (2026-10-22)
UC1 Create and Generate a Localization Project	Implemented and tested
UC2 Upload and Analyze Source Video	Implemented; faster-whisper ran on real clips; no diarization
UC3 Context-Aware Localization Preview	Implemented; Gemini candidate partial; quality not measured
UC4 Generate Localized Preview	Stock voice real (MMS-TTS); cloned voice MOCK
UC5 Review and Regenerate a Scene	Implemented and tested
UC6 Approve and Export Localization Package	Implemented and tested
UC7 Revoke Consent and Stop Identity	Implemented and tested with MOCK voices
UC8 Administer Users, Providers and Policies	Implemented and tested
UC9 Inspect Quality, Audit and Evidence	Implemented and tested

Not measured yet: transcription WER, translation quality (MQM, chrF), speaker similarity, speed, and whether context helps.

Limitations and future work

Limitations

- Formal evaluation not run; one real clip so far
- Cloned voice is MOCK; no faces or lip-sync
- No speaker diarization or visual context
- Number check compares counts, not positions
- Free tier ≈ 20 requests/day; Docker not yet run

Future work

- Run the spike: consented speakers, Tamil raters, MQM, WER, listening tests
- Context ablation vs an IndicTrans2 baseline
- Choose a cloned-voice provider from measured results
- Re-measure timing after ADR 0022
- Diarization, visual context, more language pairs

Lip-sync and face processing stay future scope: they would need their own consent design.

Thank you

Questions?

HyperLocalize turns commodity AI models into an accountable workflow: drafts are checked, humans decide, consent is enforced, and every export carries its own evidence.